

Received 20 October 2025; revised 18 November 2025; accepted 19 November 2025. Date of publication 1 December 2025; date of current version 10 December 2025. The review of this paper was arranged by Associate Editor Jacob Scharcanski.

Digital Object Identifier 10.1109/OJIM.2025.3638922

Learned-Weight Ensemble Monte Carlo DropBlock for Uncertainty Estimation and EEG Classification

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ABSTRACT Reliable predictive uncertainty—in particular predictive entropy used at decision time—in motor imagery electroencephalography (EEG)-based brain–computer interfaces (BCIs) is critical for safe real-world operation. Although Monte Carlo dropout and deep ensembles are effective for modeling predictive uncertainty, structured regularization approaches—such as FT-DropBlock for EEG-targeted convolutional neural networks (CNNs)—remain insufficiently investigated. We present learned-weight ensemble Monte Carlo FT-DropBlock with per-model temperature (LEMC-FTDB), a model-agnostic generalizable framework that integrates structured regularization, stochastic inference, learned aggregation, and per-model temperature scaling to deliver calibrated probabilities and informative uncertainty estimates on CNN models. We evaluated our framework on the EEGNet and EEG-ITNet backbones using the BCI Competition IV 2a dataset, with metrics including predictive entropy, mutual information, expected calibration error, negative log likelihood, Brier score, and misclassification-detection area under the curve (AUC). Results show that LEMC-FTDB consistently improves probability quality and calibration, strengthens misclassification detection, and maintains superior correctness and Cohen’s kappa versus Monte Carlo-based, deep ensemble-based, and deterministic baselines. Crucially, predictive entropy cleanly ranks trial reliability and enables an hold/reject policy that achieves lower risk at the same coverage in risk–coverage analyzes, supporting practical deployment. Our code is available at: github.com/SedCore/lemc_ftdb.

INDEX TERMS Brain-computer interface (BCI), deep learning (DL), digital signal processing (DSP), electroencephalography (EEG), Monte Carlo (MC) methods, uncertainty quantification (UQ).

I. INTRODUCTION

IN THE recent years, electroencephalography (EEG)-based brain–computer interfaces (BCIs) have increasingly been investigated for reliable decoding of neuro-electric brain activity. Adequate decoding of EEG is particularly challenging due to their inherent significant variance and noise, as well as the often limited size of available datasets. Variance in EEG signals stems from two main sources: 1) the uncertainty due to sensors inaccuracies or other technical limitations which mainly contributes to homoscedastic noise and 2) the intrinsic biological erratic nature of the brain activity which mainly causes the heteroscedastic data noise [1], [2]. Given the high variability in EEG signals

across subjects and sessions, it is important for BCI systems to assess the reliability of a prediction. Consequently, recent work on EEG-based deep learning (DL) models has integrated uncertainty quantification (UQ) techniques for more trustworthy predictions.

Convolutional neural networks (CNNs) have become standard for EEG classification, consistently outperforming classical machine learning and other neural network methods [3]. However, traditional neural networks provide point estimates in the form of softmax probabilities without quantifying uncertainty, which can lead to overconfident predictions on noisy or out-of-distribution EEG trials, and fail to express meaningful uncertainty when dealing with

EEG signals [4]. Hence, despite steady gains in decoding accuracy, an EEG-based BCI face a trust gap: predictions can be confidently wrong and poorly calibrated. Closing this gap requires uncertainty estimation that is both accurate and actionable.

Bayesian approaches address the deterministic nature of standard neural networks by estimating a posterior distribution over model parameters. However, exact Bayesian inference is intractable for deep neural networks [5]. Approximate techniques such as Monte Carlo dropout (MCD) [6] and deep ensembles, [7] have emerged as practical UQ alternatives. Nevertheless, dropout applies independent node-wise masking, which fails to account for spatial correlations in feature maps, particularly in CNNs. To address this, several structured regularization methods have been proposed, such as Spatial Dropout [8] and DropBlock [9]. In the context of EEG classification, FT-DropBlock has been proposed as an adaptation of DropBlock that reconsiders the shape of the input tensor at the regularization stage to align with the spatiotemporal nature of EEG signals [10].

The replacement of dropout with DropBlock during Monte Carlo (MC) inferences in an MC DropBlock (MCDB) framework improved generalization, calibration and uncertainty estimation in vision tasks [11]. The deterministic UQ (DUQ) has been applied to vision tasks, to predict classes by comparing model-extracted features to class centroids using a distance-based radial basis function (RBF) network—uncertainty is measured by the distance between the output and the nearest centroid [12]. The spectral-normalized neural Gaussian process (SNGP) extends UQ beyond vision tasks to include language processing tasks, by applying spectral normalization to hidden residual layers and replacing the output layer with a Gaussian Process, thus resulting in a hybrid deterministic–probabilistic approach [13]. However, there is currently very limited research on the application of these methods to EEG decoding tasks.

This article introduces learned-weight ensemble MC FT-DropBlock with per-model temperature (LEMC-FTDB), a modular framework for high-fidelity uncertainty estimation in EEG classification. LEMC-FTDB combines: FT-DropBlock to respect the spatiotemporal structure of EEG feature maps, MC sampling across an ensemble of CNNs to capture aleatoric uncertainty, per-model temperature scaling to amplify informative diversity among models, and a lightweight learned-weight aggregation network to optimally fuse all stochastic predictions. Together, these components deliver robust UQ and high classification accuracy while remaining agnostic to the underlying CNN backbone, thereby improving the reliability of EEG-based BCI systems by leveraging UQ for safe BCI operation.

The main contributions of this work are as follows.

- 1) A model-agnostic uncertainty framework that couples structured MC masking via FT-DropBlock with learned per-model temperatures and a tiny aggregator, thereby exploiting diversity without modifying the backbone.

- 2) Per-model temperature scaling to improve probability calibration and maintain intermodel diversity.
- 3) Data-driven aggregation through a tiny network that learns sample-wise weights from per-pass confidence or entropy, yielding sharper predictive distributions and faithful uncertainty scores than uniform averaging.
- 4) LEMC-FTDB yields consistent gains compared to baselines on probabilistic quality, calibration, and misclassification detection, with good risk–coverage tradeoffs.

The remainder of this article is organized as follows. Section II settles the background on UQ in EEG classification. Section III describes our proposed framework. Section IV details the experimental setup. Section V presents the results and discusses them. Finally, Section VI concludes this work and proposes directions for future work.

II. UNCERTAINTY QUANTIFICATION IN EEG DECODING

DL models are susceptible to uncertainties arising from inherent data noise (aleatoric uncertainty) and limitations in the training data or model structure (epistemic uncertainty) [14], [15]. Quantifying both types of uncertainty are crucial to build trustworthy systems with informed decision-making, particularly in clinical or autonomous systems where unreliable predictions can have severe consequences [16]. While traditional EEG analysis has focused on noise patterns in the source signal, the uncertainty introduced during the neural decoding process by DL models is an active area of research. Reliable uncertainty estimates in EEG decoding are important for applications like motor imagery BCI, where understanding model confidence can help mitigate hazardous movements in controlled robotic systems [1].

Various methods have been explored for UQ in DL, broadly categorized into Bayesian approaches, ensemble methods and sample-density aware techniques. Bayesian neural networks (BNNs) aim to model the posterior distribution of model weights to capture epistemic uncertainty. However, exact Bayesian inference in deep networks is reputed computationally intractable and does not have an analytic closed-form solution, thereby leading to the development of approximation techniques [17], [18].

MC dropout (MCD) is a widely used technique that approximates Bayesian inference by keeping dropout layers active during inference and performing multiple forward passes. The variance of predictions across these passes provides an estimate of epistemic uncertainty. MCD is computationally appealing as it can be applied to existing models with dropout layers. It has been applied to EEG classification tasks to provide uncertainty estimates and improve classification performance, mostly in epileptic seizure prediction [19], [20], [21], [22], while limited research applied MCD on motor imagery decoding [23], [24]. However, standard MCD is less effective in CNNs due to the dropping of individual neurons performed by dropout without considering the spatial correlation of features, resulting in poorly-calibrated uncertainty estimates [25]. MC spatial dropout (MCSD) has

recently been proposed in the context of spintronics devices as a spatial dropout-based method for structured Bayesian NN approximation considering spatial information [26]. On the other hand, MCDB extends the MCD concept by applying DropBlock during inference to estimate uncertainty in CNNs. This approach has demonstrated effectiveness in modeling uncertainty for vision tasks like object detection and segmentation [11], [25]. However, the use of MCDB in the context of EEG classification or other biosignals remains largely underexplored in the existing literature.

Deep ensembles involve training multiple models independently and aggregate their predictions to improve robustness and calibration. This UQ technique can capture epistemic uncertainty without explicitly using Bayesian formalism and often outperform MCD in practice [7]. The diversity among ensemble members contributes to better approximation of the true posterior distribution and more reliable uncertainty estimates. Deep ensembles have been applied to EEG classification, demonstrating improved performance and uncertainty estimation over MCD and similar methods [24], [27].

DUQ is a deterministic approach that fundamentally differs from Bayesian methods, as it estimates uncertainty in a single forward pass. However, a key limitation is that it does not clearly separate aleatoric and epistemic uncertainties, even though both are implicitly captured in the predicted uncertainty. While DUQ has shown superior performance compared to deep ensembles in vision tasks, some studies suggest it may underperform relative to Bayesian approaches in EEG decoding tasks with UQ [12], [24].

Recent research has explored combinations of UQ methods to leverage their complementary strengths. The combination of MCD with ensemble methods for EEG motor imagery decoding and uncertainty estimation yielded improved results than using each one separately [23]. However, the integration of ensemble methods with structured regularization-based MC simulations for UQ in EEG applications remains underexplored, representing a noticeable gap in the literature. Given the unique challenges of EEG data—including high dimensionality, low signal-to-noise ratio, nonstationarity, and intersubject variability—a comprehensive UQ approach that leverages both ensemble diversity and structured regularization could substantially advance the field.

III. PROPOSED FRAMEWORK

In this section, we describe in detail our LEMC-FTDB pipeline for high-fidelity uncertainty estimation in EEG classification. LEMC-FTDB consists of four successive stages. First, FT-DropBlock regularization preserves EEG feature coherence by imposing structured spatiotemporal sparsity. Second, MC sampling across an ensemble of M models to capture epistemic uncertainty. Third, per-model temperature scaling to learn a distinct temperature for each network and maximize informative intermodel diversity. Finally, a lightweight learned-weight multilayer perceptron

(MLP) that ingests per-pass confidence and entropy features and adaptively fuses all stochastic predictions across the M models via a composite negative log likelihood (NLL)–mutual information (MI) loss. The full architecture of our framework is presented in Fig. 1.

A. FT-DROPBLOCK FOR BAYESIAN APPROXIMATION

Conventional Dropout drops individual activations at random, ignoring the fact that EEG feature maps exhibit strong correlations across both feature (channel-filter) and time dimensions. FT-DropBlock [10] adapts the DropBlock regularization to EEG by dropping contiguous blocks along the channel-filter and time axes of a convolutional feature map within single-model CNNs. In the MC inference setting of the present work, each instantiation of the binary mask $\mathcal{M}$ at inference effectively constitutes a sample from a posterior distribution over the intermediate activations of the network—analogue to sampling posterior weight configurations under dropout. Because FT-DropBlock zeros out contiguous blocks of feature-time activations rather than dropping individual activation units, the network is forced to learn more robust representations that do not rely on narrow time-feature patterns, leading to richer variations in the predictive distribution across MC samples.

By keeping FT-DropBlock active during both training and test time (i.e., by sampling new structured masks $\mathcal{M}$ for each forward pass—the network performs a form of MC estimate of its predictive distribution. Concretely, for a test trial i , each FT-DropBlock instantiation yields a different activation pathway and, hence, a different softmax output

$$p_i^{(t)} = \text{softmax}(f_{\theta, \mathcal{M}^{(t)}}(x_i)) \in \mathbb{R}^K, \quad t = 1, \dots, T \quad (1)$$

where $f_{\theta, \mathcal{M}^{(t)}}$ denotes the network with mask $\mathcal{M}^{(t)}$ applied. Averaging these T stochastic draws approximates the posterior predictive

$$\bar{p}_i \approx \mathbb{E}_M[\text{softmax}(f_{\theta, \mathcal{M}}(x_i))]. \quad (2)$$

B. ENSEMBLE MONTE CARLO SAMPLING

To approximate the posterior over network parameters and capture model uncertainty, we train in total M independent CNN models with FT-DropBlock active during both training and inference. During inference, each model m of the ensemble produces T stochastic forward-passes (sampling new FT-DropBlock masks), yielding per-pass probabilities

$$p_i^{(m,t)} = \text{softmax}(z_i^{(m,t)}) \in \mathbb{R}^K, \quad i = 1, \dots, N, \quad t = 1, \dots, T, \quad m = 1, \dots, M \quad (3)$$

where N is the total number of trials in the test dataset, $z_i^{(m,t)}$ is the presoftmax logit vector produced by model m (out of M) on its t th stochastic pass for the i th trial, and K is the total number of classes in the dataset.

Stacking over the total number of stochastic draws, which is the number of models M times the number of MC passes T , gives

$$\{p_j\}_{j=1}^{MT} \subset \mathbb{R}^K. \quad (4)$$

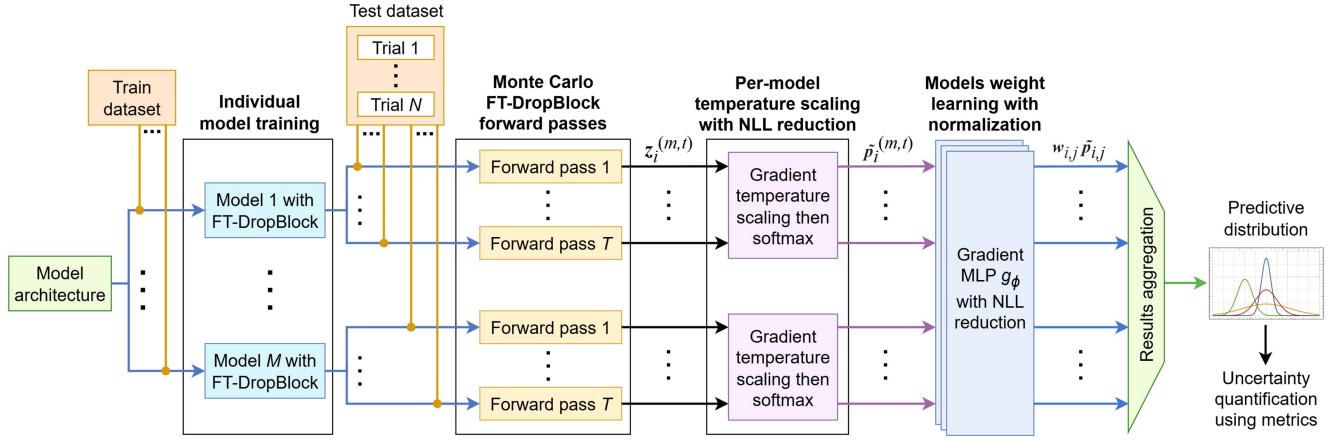


FIGURE 1. Architecture of the LEMC-FTDB framework.

The predictive mean $\bar{p}_i$ captures total uncertainty, while downstream metrics (entropy, MI) quantify epistemic components

$$\bar{p}_i = \frac{1}{MT} \sum_{j=1}^{MT} p_{i,j}. \quad (5)$$

C. PER-MODEL TEMPERATURE SCALING

We apply temperature scaling to calibrate the softmax outputs, i.e., to make predicted confidences better match true accuracies by appropriately sharpening or flattening the logits. However, uniform temperature scaling may under- or over-sharpen the outputs of some models in the ensemble. Instead, we learn a distinct temperature $\tau_m > 0$ from for each ensemble member. Using the presoftmax logits $z_i^{(m,t)}$, we form $\tilde{z}_i^{(m,t)}$ and $\tilde{p}_i^{(m,t)}$ which are, respectively, the scaled logits and the scaled probabilities

$$\tilde{z}_i^{(m,t)} = \frac{z_i^{(m,t)}}{\tau_m} \quad (6)$$

$$\tilde{p}_i^{(m,t)} = \text{softmax}(\tilde{z}_i^{(m,t)}). \quad (7)$$

Then, we minimize the NLL $\mathcal{L}_{\text{NLL}}$ of the ensembled mean

$$\mathcal{L}_{\text{NLL}}(\{\tau_m\}) = -\frac{1}{N} \sum_{i=1}^N \log \left(\frac{1}{MT} \sum_{m,t} \tilde{p}_{i,y_i}^{(m,t)} \right) \quad (8)$$

with respect to $\{\tau_m\}$, clipping each $\tau_m \in [0.1, 10]$, with $y_i \in \{1, \dots, K\}$ being the ground-truth class label for the i th EEG trial. This per-model scaling boosts MI and improves calibration.

Minimizing the NLL of the ensembled mean implements the temperature-scaling principle by tuning each τ_m so that the final averaged predictive distribution $\bar{p}_i$ best matches the true labels in the log-likelihood sense. This is achieved through calibration and discrimination. The resulting temperature settings $\{\tau_m\}$ yields the best-calibrated, highest-likelihood ensemble predictions.

Calibration consists in adjusting the sharpness of the outputs of each model, so that, once averaged, the predicted

probabilities are neither systematically over- nor under-confident. Discrimination is achieved by pushing down the loss on misclassified examples, thereby making the ensemble mean more peaked on the correct class, improving overall accuracy and log-score.

Each ensemble member m learns its own temperature τ_m so that its softmax outputs are neither too sharp nor too flat. A model whose raw logits are overly peaky (overconfident) will be assigned a larger τ_m , which flattens its output distribution and increases its entropy. Conversely, a model whose logits are too diffuse (underconfident) will receive a smaller τ_m , which sharpens its predictions. By learning $\{\tau_m\}$ to minimize the ensembled NLL, we automatically rebalance the confidence of each network: overconfident members are toned down, underconfident ones are amplified, and the final average better matches the true label distribution.

D. LEARNED-WEIGHT AGGREGATION

At this last step, we fuse the $M \times T$ scaled probabilities $\{\tilde{p}_{i,j}\}$ with input-dependent weights. For each test sample i of class k and MC pass j , we compute two scalar features for the pass: 1) its confidence $c_{i,j}$ which represents its highest softmax probability and 2) its entropy $h_{i,j}$, as follows:

$$c_{i,j} = \max_k \tilde{p}_{i,j,k} \quad (9)$$

$$h_{i,j} = -\sum_k \tilde{p}_{i,j,k} \log(\tilde{p}_{i,j,k}). \quad (10)$$

Then, we stack these two scalars into a 2-D feature vector $\mathbf{x}_{i,j} = (c_{i,j}, h_{i,j}) \in \mathbb{R}^2$ that we feed into a small MLP $g_\phi(\mathbf{x})$. The MLP acts as a tiny feed-forward network composed of two hidden layers—first with 16 ReLU-activated units, then 8 ReLU-activated units—followed by a single-neuron output layer with a softplus activation to guarantee strictly positive aggregation weights

$$g_\phi(\mathbf{x}) = \text{softplus}(W_2 \text{ReLU}(W_1 \mathbf{x} + b_1) + b_2) \in \mathbb{R}_{>0} \quad (11)$$

where W_1 and W_2 are, respectively, the weight matrix of the first and the second hidden layer, b_1 and b_2 are bias vector

of the first and the second layer, respectively. These four parameters are learned jointly to produce adaptive, positive aggregation weights for each stochastic pass.

Each $\mathbf{x}_{i,j}$ goes through these layers to yield a raw weight $\hat{w}_{i,j} = g_\phi(\mathbf{x}_{i,j}) \geq 0$. Then, we normalize those $\hat{w}$ across all j for sample i to get our final aggregation weights used to form the weighted average of the $M \times T$ predictions

$$w_{i,j} = \frac{\hat{w}_{i,j}}{\sum_{j'} \hat{w}_{i,j'}}. \quad (12)$$

The final predictive distribution is then given by

$$\hat{p}_i = \sum_{j=1}^{MT} w_{i,j} \tilde{p}_{i,j} \quad (13)$$

where the weights satisfy $w_{i,j} \geq 0$, $\sum_j w_{i,j} = 1$. Finally, we train the aggregation network g_ϕ

$$\mathcal{L}(\phi) = -\frac{1}{N} \sum_{i=1}^N \log \hat{p}_{i,y_i} \quad (14)$$

where $\hat{p}_{i,y_i}$ is the learned-weight average of the MT stochastic predictions $\tilde{p}_{i,j}$. The aggregation network g_ϕ is trained by minimizing the NLL of the learned-weight ensemble mean $\hat{p}_i$. This directly rewards placing high probability on the true class while letting the weights $w_{i,j}$ emphasize the most informative stochastic passes.

Trained solely with NLL, g_ϕ learns a simple rule: up-weight stochastic passes that are confident and low-entropy (calibrated, consensus-supporting) and down-weight uncertain or overconfident outliers. Because weights are normalized across passes, no single spike can dominate; probability mass is reallocated toward calibrated evidence, which empirically lowers NLL/expected calibration error (ECE) without sacrificing correctness.

IV. EXPERIMENT SETUP

A. DATASET

We use the BCI Competition IV 2a dataset (BCIC-IV 2a) [28] in this study, from the mother of All BCI benchmarks (MOABBs) library version 1.2.0 [29]. This dataset contains multiclass motor imagery EEG data from 9 healthy Subjects, recorded in two sessions on different days. Each session consisted of 6 runs separated by short breaks to prevent Subjects from accumulating mental fatigue. Each run comprises 12 trials of each of the 4 classes, respectively, Left hand (class 1), Right hand (class 2), Feet (class 3), and Tongue (class 4). Each trial was recorded during 7.5 s. EEG signals were recorded from 22 electrodes at 250 Hz, and bandpass-filtered between 0.5 Hz and 100 Hz. A 50 Hz notch filter was used to remove power line interference.

In our experiment, we use the BCIC-IV 2a dataset at its default sampling frequency of 250 Hz, with a time window length of 4.5 s, starting 0.5 s before the cue until the end of the motor imagery task. We applied standardization on the EEG signals by using an exponential moving average

window with a decay factor of 0.999 as in [30]. We do not perform additional bandpass-filtering on the EEG signals as this has proven to yield better overall results when working with this dataset [10], [31].

B. CONVOLUTIONAL NEURAL NETWORKS

1) EEGNET

EEGNet [30] is a compact CNN for EEG-based BCIs which has been widely adopted in motor imagery EEG classification for its simple and clear design, making it a foundational model on the top of which have been built at least 14 other CNN architectures for EEG decoding [32]. In this study, we used the same model parameters as in [30]: $F_1 = 8$ temporal filters, $D = 2$ spatial filters, $F_2 = F_1 \times D = 16$ point-wise filters, and temporal kernel length of 32.

2) EEG-ITNET

EEG-ITNet [33] is an inception temporal convolutional network designed for motor imagery decoding. It uses inception modules and causal convolutions with dilatation to extract rich spectral, spatial, and temporal information from multichannel EEG data with few trainable parameters. In this study, we used the model with the same parameters as in [33]: 2, 4, and 8 filters; kernel sizes of 16, 32, and 64 samples, respectively, in the three 2-D convolutional layers.

In both architectures, we replaced all dropout layers with FT-DropBlock layers. We trained each ensemble member for 500 epochs with a batch size of 64, an Adam optimizer at a learning rate set to 0.001, a drop probability of $p = 0.5$, and a block size set to $c = 15$ as in [10].

C. MODEL PERFORMANCE METRICS

Overall model Correctness (A)—commonly referred to as Accuracy in machine learning—is the ratio of the number of correct predictions over the total number of predictions. The Cohen's kappa score κ measures inter-rater prediction agreement, correcting for agreement that could occur by chance

$$A = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} \quad (15)$$

$$\kappa = \frac{p_o - p_e}{1 - p_e} \quad (16)$$

where p_o is the observed accuracy and p_e the expected chance agreement. For both metrics, higher values are better.

D. UNCERTAINTY EVALUATION METRICS

1) PREDICTIVE ENTROPY

Predictive entropy $\mathbb{H}_{\text{pred}}$ measures the total uncertainty associated with the output of a model by computing the Shannon entropy of the averaged predictive distribution [34], [35]. Formally, for a sample x , it is defined as

$$\mathbb{H}_{\text{pred}}(x) = -\sum_{k=1}^C \bar{p}_k(x) \log \bar{p}_k(x) \quad (17)$$

where C is the total number of classes, and $\bar{p}_k(x)$ is the mean predicted probability for class k across all MC samples and ensemble members. This metric captures the total uncertainty, i.e., both epistemic and aleatoric uncertainty [36]. High entropy values indicate uncertain predictions, often due to class ambiguity or out-of-distribution inputs.

2) MUTUAL INFORMATION

MI isolates the epistemic component of uncertainty, which is due to lack of model knowledge [37]. It is the difference between the predictive entropy and the expected entropy $\mathbb{E}_t$ of individual stochastic predictions

$$\mathbb{I}(x) = \mathbb{H}_{\text{pred}}(x) - \mathbb{E}_t[\mathbb{H}(p^{(t)}(x))]. \quad (18)$$

High $\mathbb{I}(x)$ indicates that the model exhibits disagreement across stochastic passes, reflecting uncertainty about the correct label due to insufficient training data or model capacity, meaning the model is aware that it does not know.

3) EXPECTED CALIBRATION ERROR

While entropy, MI, and variance quantify uncertainty, ECE assesses how well predicted probabilities align with actual outcomes [38]. ECE bins predictions by confidence and computes the weighted average of the absolute difference between average confidence and empirical accuracy in each bin

$$\text{ECE} = \sum_{b=1}^B \frac{|B_b|}{n} |\text{acc}(B_b) - \text{conf}(B_b)| \quad (19)$$

where n is the total number of samples, B_b is the set of samples in bin b , $\text{acc}(B_b)$ is the accuracy within the bin, and $\text{conf}(B_b)$ is the average confidence. A low ECE indicates that the predicted probabilities of the model are well-calibrated.

4) NEGATIVE LOG-LIKELIHOOD

The NLL (also called the cross-entropy loss on held-out data) is a commonly-used metric to assess the quality of epistemic uncertainty on a held-out dataset [39]. For a set of N samples $\{(x_i, y_i)\}$, let $\bar{p}_i = [\bar{p}_{i,1}, \dots, \bar{p}_{i,K}]$ be the predictive mean probability vector of the model over K classes. The NLL is defined as

$$\text{NLL} = -\frac{1}{N} \sum_{i=1}^N \log(\bar{p}_{i,y_i}) \quad (20)$$

where $\bar{p}_{i,y_i}$ is the probability assigned to the true class. NLL penalizes confident wrong predictions sharply. It combines calibration and accuracy: a well-calibrated model will assign high probability to the correct class, yielding low NLL.

5) BRIER SCORE

The Brier score for multiclass classification [40] is the mean squared error between the one-hot label vector $y_i^{(\text{one-hot})} \in \{0, 1\}^K$ and the predictive mean $\bar{p}_i$

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N \sum_{k=1}^K (\bar{p}_{i,k} - y_{i,k}^{(\text{one-hot})})^2 \quad (21)$$

The Brier score measures the squared-error between predicted probabilities and true labels. It is sensitive both to calibration (whether probabilities match frequencies) and sharpness (how peaked the distribution is). Lower scores are better.

6) MISCLASSIFICATION-DETECTION AREA UNDER THE CURVE

Area under the curve (AUC) is a robust metric to evaluate the predictive ability of learning algorithms [41]. We define the misclassification-detection AUC [AUC receiver operating characteristic (ROC)] as the area under the ROC curve obtained by treating the predictive entropy of sample i , denoted H_i . AUC ROC acts as a detection score for the binary error indicator $\delta_i = \mathbf{1}\{\arg \max_k \bar{p}_{i,k} \neq y_i\}$ which is 1 if the model misclassifies x_i , 0 otherwise.

The AUC ROC indicates how well entropy ranks misclassifications above correct predictions. An AUC ROC of 1.0 means all wrong predictions have strictly higher entropy than all correct ones. Hence, AUC ROC values closer to 1.0 are better.

E. UNCERTAINTY EVALUATION METHODS

In this work, we compare five state-of-the-art methods: single-model MCD [6], single-model MCSD [26], single-model MC FT-DropBlock (MC-FTDB) [11], Deep Ensemble with dropout active only during model training (E-Dropout) [24], Deep Ensemble MCD with dropout active during both training and testing (EMCD) [23], SNGP [13], and our proposed LEMC-FTDB. Table 1 summarizes the uncertainty evaluation methods we compare in this study.

For the DUQ method, we removed all stochastic layers (i.e., dropout) from CNN backbones to keep the network deterministic. The final dense activation layer has been replaced with a dense layer of 100 units using an ReLU activation, followed by an RBF classification layer with a length scale of 0.5 and trainable centroids of dimension 100. L2-regularization with a strength set to 0.01 was then applied to trainable weights in all layers of the obtained DUQ models, which were compiled using binary cross-entropy and an Adam optimizer with a learning rate set to 0.01 [12], [24]. Since DUQ is deterministic and provides distance-based confidence in a single deterministic forward pass rather than a probabilistic posterior over parameters, MI and predictive variance do not apply as they require a distribution over model predictions.

In the SNGP method, the final random feature Gaussian process layer returns the mean and covariance matrix of the predictive Gaussian distribution of logits. We applied the Softmax function on the obtained predictive mean of logits to get the final probabilities used to compute A and κ , and we extracted the predictive variance from the diagonal of the covariance matrix. The remaining metrics were calculated by sampling $T = 200$ logits from the Gaussian distribution and applying the Softmax function to convert them into predicted probabilities. Empirically,

TABLE 1. Uncertainty evaluation methods under comparison in this study.

Method	Stack type	Changes to model architecture	Multiple model training	Stochastic inferences
Monte Carlo Dropout (MCD) [6]	Single-model	No	No	Yes
Monte Carlo Spatial Dropout (MCSD) [26]	Single-model	Yes: Replaces Dropout layers with Spatial Dropout layers	No	Yes
Monte Carlo FT-DropBlock (MC-FTDB) [11]	Single-model	Yes: Replaces Dropout layers with FT-Dropblock layers	No	Yes
Deep Ensemble (E-Dropout) [24]	Ensemble	No	Yes	No
Deep Ensemble Monte Carlo Dropout (EMCD) [23]	Ensemble	No	Yes	Yes
Deterministic Uncertainty Quantification (DUQ) [12]	Single-model	Yes: Removes stochastic layers and replaces the output layer with a RBF	No	No
Spectral-normalized Neural Gaussian Process (SNGP) [13]	Single-model	Yes: Applies spectral normalization to hidden layers and replaces the output layer with a Gaussian process	No	No
Learned-Weight Ensemble Monte Carlo FT-DropBlock with Per-Model Temperature (LEMC-FTDB)	Ensemble	Yes: Replaces Dropout layers with FT-Dropblock layers	Yes	Yes

we found that using $512 \times 3 = 1536$ and 512 random Fourier features to approximate the Gaussian process yielded the best results for EEGNet and EEG-ITNet, respectively. All other parameters were kept unchanged from the Tensorflow Model Garden v2.19.1 implementation of SNGP [42].

The per-model temperature and the MLP of our LEMC-FTDB were trained, respectively, for 500 steps and 200 epochs, both with an Adam optimizer at a learning rate of 0.001. We ran our experiment on a computer equipped with an NVIDIA GeForce RTX 2080 GPU, running Ubuntu 24.04 operating system and the CUDA 12.9 library. To ensure reproducibility, our code is publicly available at: github.com/SedCore/lemc_ftdb.

V. RESULTS AND DISCUSSION

We used the BCIC-IV 2a dataset in a hold-out strategy for each of the nine subjects: the subdataset of the first EEG recording day used for model training, and the subdataset of the second day for model testing. Each subdataset comprises 72 trials per class, i.e., 288 trials in total for all 4 classes. We retain all trials (including potential artifacts) to evaluate model robustness under realistic EEG noise.

For each of the CNN architectures used in this study, we formed an ensemble of $M = 15$ models by training independent replicas with different random initializations on the same subject-specific train and test splits. During training, the training set is shuffled each epoch (default Tensorflow Keras behavior), but no data resplitting or data augmentation is used. All hyperparameters are identical across ensemble members. For each ensemble member, we conducted $T = 200$ passes during MC inferences. We empirically set the total number of bins to $B = 10$ in the evaluation of the ECE. As a matter of preference, we computed the Shannon entropy in bits.

A. MODEL PERFORMANCE

Tables 2 and 3 report model performance for EEGNet and EEG-ITNet. The standard deviation (SD) summarizes between-subject variability of model correctness. The 95% confidence interval (CI) of the mean is computed with a Student t distribution ($df=8$). It gives the precision of the mean across the nine subjects, indicating the range in which the true population mean would fall 95% of the time under repeated sampling.

We observe that across both architectures, LEMC-FTDB consistently achieves the highest correctness ($A\%$) and inter-rater agreement with ground truth beyond chance (κ), with competitive stability (SD and 95% CI) across subjects. With EEGNet, LEMC-FTDB attains 80.40% correctness and $\kappa = 0.74$ over 9 subjects, exceeding the strongest non-LEMC baselines (MC-FTDB and EMCD) by roughly +5 percentage points (pp) in correctness and +0.06 in κ . With EEG-ITNet, LEMC-FTDB again ranks first with 82.48% and $\kappa = 0.77$, thereby outperforming EMCD and MC-FTDB by roughly +5 pp and +0.06 κ .

Across both backbones, the dispersion of LEMC-FTDB is moderate with smaller SD than MC-FTDB/EMCD, and comparable to the most stable but less accurate MCSD. This denotes a more consistent behavior across the cohort and smaller intersubject performance variability, indicating its gains are not driven by a single outlier subject. The tight 95% confidence t -intervals around the mean observed with both backbones indicates reasonably precise estimates given the low number of subjects $n = 9$.

The deterministic DUQ and the hybrid SNGP trail both in mean accuracy and κ . Overall, the results show consistent cross-subject gains for LEMC-FTDB, with uncertainty bands that are comparable to or tighter than most baselines. While CIs across methods still overlap (as expected with $n = 9$), higher means of LEMC-FTDB in both correctness and κ , combined with comparable or tighter SD and CIs, point to

TABLE 2. Comparison of classification correctness (%) and kappa score across methods using EEGNet.

Subjects	MCD		MCSD		MC-FTDB		E-Dropout		EMCD		DUQ		SNGP		LEMC-FTDB	
	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ
S1	81.94	0.76	78.82	0.72	83.33	0.78	81.94	0.76	83.33	0.78	72.92	0.64	79.86	0.73	86.46	0.82
S2	59.38	0.46	60.07	0.47	63.89	0.52	60.42	0.47	63.19	0.51	49.31	0.32	52.78	0.37	69.44	0.59
S3	84.72	0.80	86.46	0.82	91.67	0.89	89.58	0.86	90.97	0.88	86.81	0.82	87.15	0.83	92.01	0.89
S4	62.15	0.50	61.81	0.49	69.44	0.59	64.93	0.53	64.24	0.52	63.19	0.51	63.19	0.51	73.96	0.65
S5	68.75	0.58	67.71	0.57	70.49	0.61	71.53	0.62	73.61	0.65	69.79	0.60	68.06	0.57	74.65	0.66
S6	57.64	0.44	59.72	0.46	55.21	0.40	58.68	0.45	56.60	0.42	57.99	0.44	55.56	0.41	63.19	0.51
S7	72.57	0.63	80.21	0.74	89.58	0.86	85.07	0.80	86.11	0.81	73.26	0.64	75.35	0.67	91.32	0.88
S8	79.51	0.73	66.67	0.56	78.82	0.72	78.12	0.71	80.21	0.74	72.22	0.63	68.40	0.58	86.81	0.82
S9	73.61	0.65	73.96	0.65	81.25	0.75	79.17	0.72	79.86	0.73	71.53	0.62	76.39	0.69	85.76	0.81
Mean	71.14	0.62	70.60	0.61	75.96	0.68	74.38	0.66	75.35	0.67	68.56	0.58	69.60	0.59	80.40	0.74
SD	9.91	0.13	9.70	0.13	12.10	0.16	11.06	0.15	11.69	0.16	10.66	0.14	11.34	0.15	10.32	0.14
95 % CI	7.62	0.10	7.46	0.10	9.30	0.12	8.50	0.11	8.99	0.12	8.19	0.11	8.72	0.12	7.93	0.11

TABLE 3. Comparison of classification correctness (%) and kappa score across methods using EEG-ITNet.

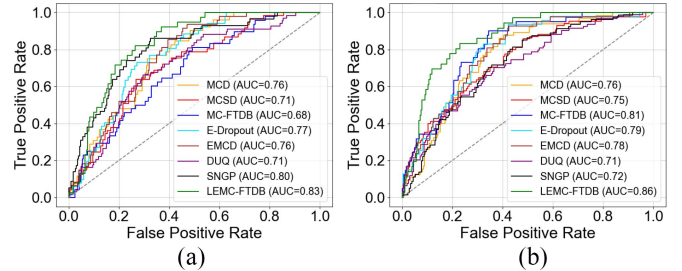
Subjects	MCD		MCSD		MC-FTDB		E-Dropout		EMCD		DUQ		SNGP		LEMC-FTDB	
	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ	A %	κ
S1	77.43	0.70	70.49	0.61	85.76	0.81	83.68	0.78	85.42	0.81	62.15	0.50	71.88	0.63	87.50	0.83
S2	60.42	0.47	55.90	0.41	63.19	0.51	64.93	0.53	64.58	0.53	47.57	0.30	54.86	0.40	70.14	0.60
S3	86.46	0.82	69.10	0.59	90.28	0.87	88.89	0.85	91.32	0.88	85.76	0.81	70.14	0.60	93.06	0.91
S4	66.32	0.55	64.93	0.53	65.97	0.55	74.65	0.66	73.61	0.65	68.40	0.58	62.15	0.50	80.56	0.74
S5	70.14	0.60	71.53	0.62	68.06	0.57	71.18	0.62	72.57	0.63	66.32	0.55	67.71	0.57	78.12	0.71
S6	56.60	0.42	55.21	0.40	59.72	0.46	60.76	0.48	61.81	0.49	57.29	0.43	52.43	0.37	68.75	0.58
S7	71.88	0.63	67.71	0.57	89.24	0.86	87.85	0.84	88.89	0.85	81.60	0.75	86.81	0.82	92.36	0.90
S8	76.39	0.69	64.24	0.52	82.99	0.77	76.39	0.69	81.94	0.76	73.61	0.65	64.93	0.53	85.76	0.81
S9	80.90	0.75	66.67	0.56	87.50	0.83	87.15	0.83	85.42	0.81	76.74	0.69	76.39	0.69	86.11	0.81
Mean	71.84	0.62	65.09	0.53	76.97	0.69	77.28	0.70	78.40	0.71	68.83	0.58	67.48	0.57	82.48	0.77
SD	9.64	0.13	5.90	0.08	12.45	0.17	10.34	0.14	10.67	0.14	12.10	0.16	10.61	0.14	8.82	0.12
95 % CI	7.41	0.10	4.54	0.06	9.57	0.13	7.95	0.11	8.20	0.11	9.30	0.12	8.15	0.11	6.78	0.09

systematic, cross-subject improvements and stronger beyond-chance agreement versus all baselines.

B. UNCERTAINTY EVALUATION

Fig. 2 shows ROC curves with their respective AUCs for Subject 1 using both backbones. In EEGNet [Fig. 2(a)], LEMC-FTDB yields the highest separability between correct and incorrect trials with $AUC = 0.83$, dominating the low-FPR region ($FPR < 0.2$) where selective rejection is most valuable. The next best are SNGP ($AUC = 0.80$) and E-Dropout ($AUC = 0.77$), while MCSD/DUQ (≈ 0.71) and MC-FTDB (0.68) lag, indicating weaker uncertainty signals. In EEG-ITNet [Fig. 2(b)], LEMC-FTDB again leads ($AUC = 0.86$), with MC-FTDB (0.81) and E-Dropout (0.79) following; deterministic methods (DUQ 0.71, SNGP 0.72) underperform. The consistent ROC dominance—especially at small FPR—suggests that the learned per-model temperatures and data-driven weighting of LEMC-FTDB produce uncertainty estimates that better distinguish hard/incorrect trials from easy/correct ones, improving practical error detection without sacrificing backbone choice.

Fig. 3 presents reliability diagrams for Subject 2. Perfect calibration would trace the diagonal; curves above (below)


FIGURE 2. ROC curves for Subject 1 with (a) EEGNet and (b) EEG-ITNet.

the diagonal indicate under- (over-) confidence. With EEGNet [Fig. 3(a)], LEMC-FTDB tracks the diagonal closely from mid to high confidence (≈ 0.5 – 0.9) and reaches near-perfect calibration in the top bin, indicating that when it is confident it is usually correct—consistent with its low ECE reported in Fig. 8. MC-FTDB and E-Dropout show similar trend; EMCD is competitive but tends to slight underconfidence in mid bins, while MCD oscillate around the diagonal with noticeable overconfidence at 0.4 – 0.6 . SNGP shows a systematic overconfidence trend, and DUQ is markedly miscalibrated, with large deviations

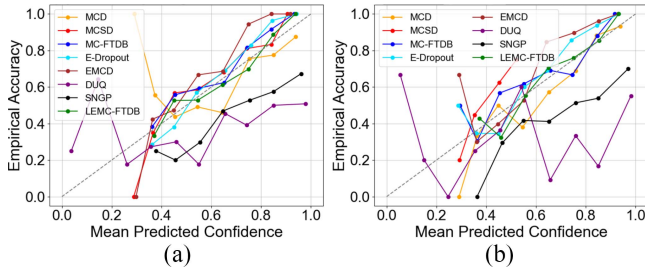


FIGURE 3. Reliability diagram for subject 2 with (a) EEGNet and (b) EEG-ITNet.

and nonmonotone behavior across bins. The pattern repeats with EEG-ITNet [Fig. 3(b)]: LEMC-FTDB is the closest to the diagonal over most bins, followed by MC-FTDB and E-Dropout. MCD improve at high confidence but remains slightly overconfident in the middle range while EMCD is underconfident in that range. SNGP stays overconfident, and DUQ remains the least calibrated. Overall, LEMC-FTDB provides the most consistent probability–accuracy alignment across confidence levels for this subject, particularly in the high-confidence regime that dominates decision making.

Fig. 4 examines predictive entropy conditioned on outcome (black = correct, red = incorrect) for Subject 3 and both backbones. A well-behaved uncertainty method should produce low entropy for correct trials and substantially higher entropy for misclassified trials, with minimal overlap between the two distributions. LEMC-FTDB best matches this behavior in both EEGNet and EEG-ITNet: the correct-trial boxes sit near the low-entropy regime with tight interquartile ranges (IQRs), while the misclassification boxes are clearly shifted upward, yielding the largest separation among methods and indicating confident decisions when right and elevated uncertainty when wrong. MC-FTDB and EMCD also show the expected ordering but with noticeably more overlap, implying less reliable awareness of incorrect trials. MCS D often inflates entropy for both outcomes (underconfidence), thereby reducing separability. DUQ on single passes exhibits very low entropy even on incorrect trials (overconfidence on mistakes), and SNGP tends to raise entropy broadly, including on correct trials (systematic underconfidence). Taken together, the conditional entropy profiles on this figure show that LEMC-FTDB provides the clearest discrimination between correct and incorrect trials, aligning uncertainty with actual misclassification events rather than uniformly shifting confidence up or down.

Fig. 5 shows risk–coverage tradeoffs for Subject 4 (lower risk at a given coverage is better). Across both backbones, the LEMC-FTDB curve (green) lies strictly below the baselines, indicating that its uncertainty scores enable a more effective reject option: for the same misclassification budget it retains more trials, and for the same coverage it yields fewer mistakes. In the low-coverage regime (left), LEMC-FTDB rapidly approaches near-zero risk—evidence that the most confident predictions are reliably correct—while other methods exhibit spikes/plateaus, revealing misranked confidence where some retained samples are still incorrect-prone.

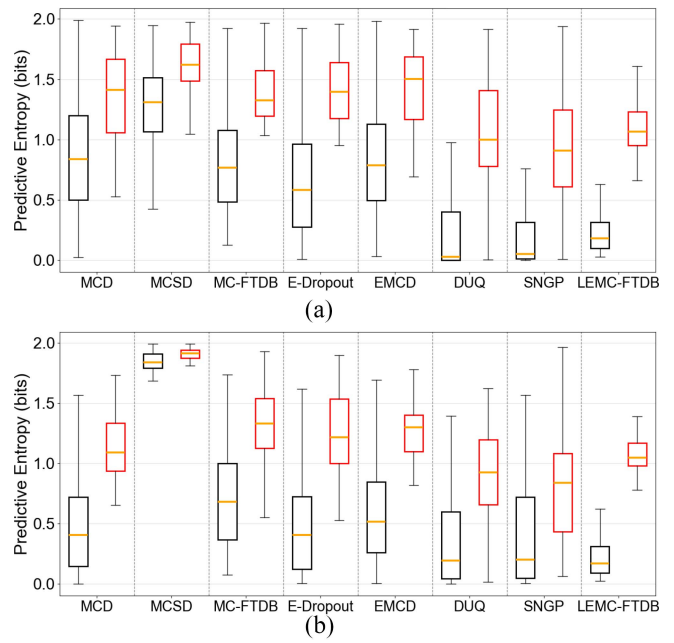


FIGURE 4. Entropy by outcome for subject 3 with (a) EEGNet and (b) EEG-ITNet.

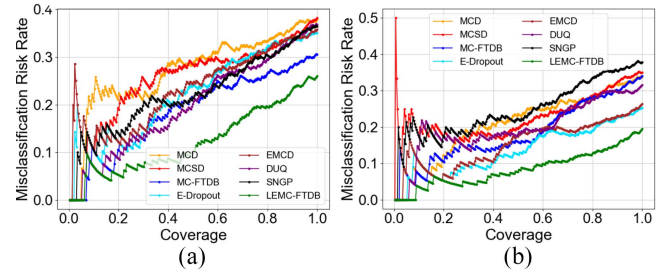


FIGURE 5. Risk–coverage curves for subject 4 with (a) EEGNet and (b) EEG-ITNet.

Overall, LEMC-FTDB maintains the lowest risk up to high coverage levels, reflecting better calibration and ranking of uncertainty rather than merely shifting confidence.

Fig. 6 shows that LEMC-FTDB yields the largest MI per subject except for S3 in both backbones, while SNGP remains low–moderate, implying limited epistemic spread in hybrid deterministic methods. DUQ remains at zero since it uses single deterministic passes. MC-FTDB and EMCD yield intermediate MI, consistent with weaker diversity control. This elevated MI coexists with superior correctness, κ , and risk–coverage, indicating that LEMC-FTDB increases epistemic diversity without sacrificing predictive quality.

Fig. 7 shows that LEMC-FTDB yields the lowest or near-lowest NLL across subjects, including the harder cases (S2 and S6) where deterministic DUQ spikes dramatically, revealing severe miscalibration. MC-FTDB, E-Dropout, and EMCD form the next tier with moderate NLL. SNGP is consistently above LEMC-FTDB, indicating less reliable probability estimates. Overall, the curves show that LEMC-FTDB provides sharper and more trustworthy predictive

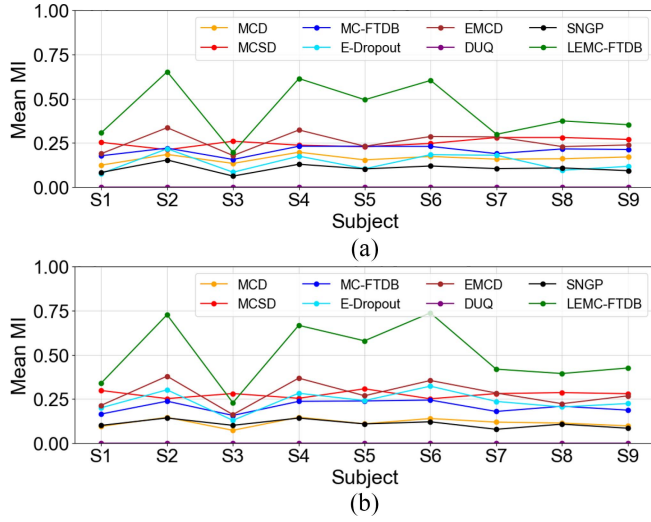


FIGURE 6. Mean MI with (a) EEGNet and (b) EEG-ITNet.

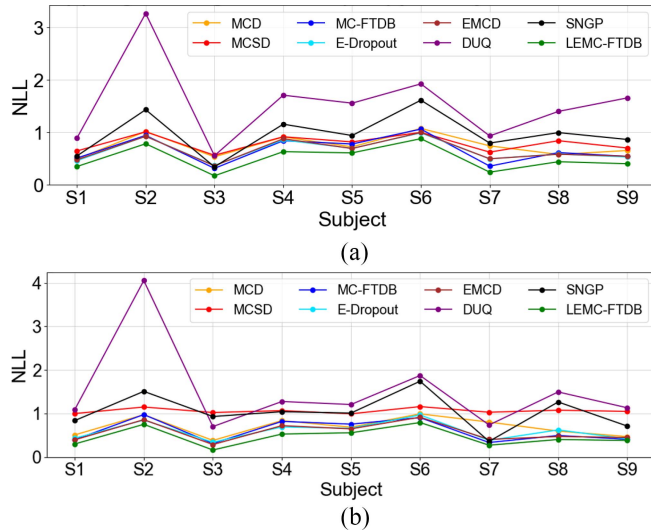


FIGURE 7. NLL with (a) EEGNet and (b) EEG-ITNet.

probabilities across individuals, while DUQ and, to a lesser extent, SNGP exhibit systematic likelihood deficits.

Fig. 8 shows that across subjects and backbones, LEMC-FTDB delivers the lowest or near-lowest ECE. On difficult subjects (S2 and S6), DUQ and SNGP exhibits large miscalibration spikes. MC-FTDB and E-Dropout form a second tier with modest ECE, while MCS and EMCD fluctuate more and occasionally overshoot. Overall, per-model temperature scaling plus learned weighting in LEMC-FTDB produces well-calibrated probabilities across subjects, whereas DUQ and SNGP tend to be systematically miscalibrated.

Fig. 9 shows that LEMC-FTDB yields the lowest or near-lowest Brier, indicating the most accurate probabilistic forecasts. MCS, DUQ, and SNGP tend to exhibit the worst results, while MC-FTDB, E-Dropout, and EMCD form a second tier with moderate Brier values. Overall, the curves show that LEMC-FTDB delivers the best combined

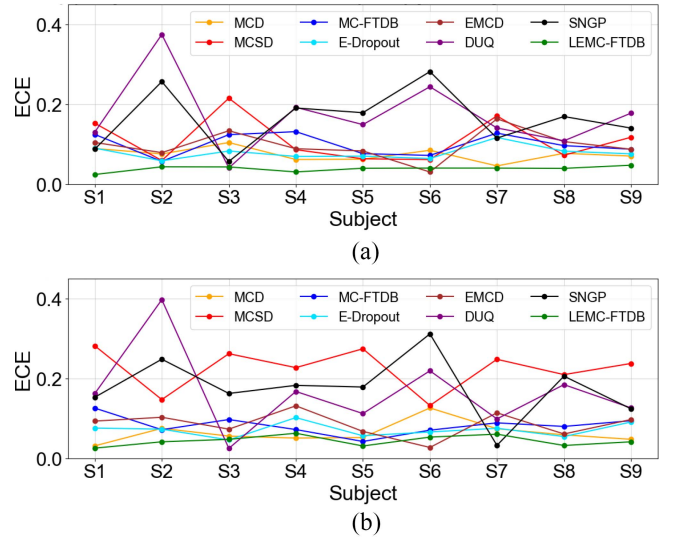


FIGURE 8. ECE with (a) EEGNet and (b) EEG-ITNet.

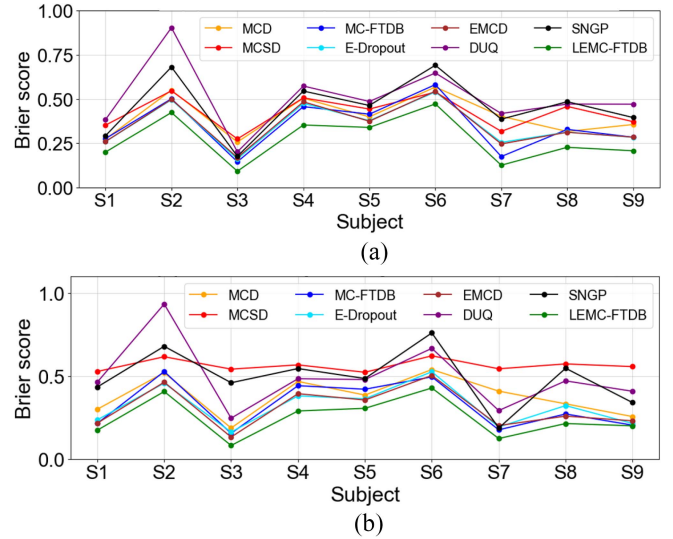


FIGURE 9. Brier score with (a) EEGNet and (b) EEG-ITNet.

calibration plus refinement, not merely high accuracy, thus yielding the most reliable probability estimates subject-wise.

ECE captures reliability (calibration) whereas the Brier score blends reliability and sharpness (overall probabilistic quality). A model that issues cautious, near-uniform probabilities may achieve low ECE (confidence matches correctness) but higher Brier due to poor sharpness. Conversely, a model with confident, mostly correct predictions can attain lower Brier even if ECE is higher (slight over- or under-confidence). In safety-critical BCI control, calibration dominates since overconfident false activations are costly. Hence, we emphasize ECE, NLL, and the shape of risk-coverage curves (slow misclassification accumulation). At the same time, LEMC-FTDB preserves useful sharpness, as reflected by lower NLL and Brier relative to baselines.

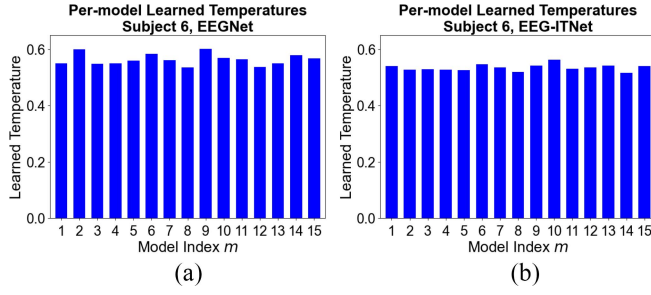


FIGURE 10. Per-model learned temperatures for subject 6 with (a) EEGNet and (b) EEG-ITNet.

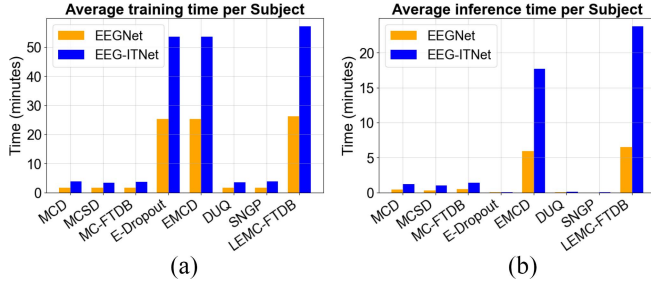


FIGURE 11. Computational times with (a) EEGNet and (b) EEG-ITNet.

Fig. 10 shows that for Subject 6, the per-model learned temperatures cluster tightly on both backbones, with most models around 0.50–0.55. Because temperature scales logits as z/τ , values below 1 indicate calibrated sharpening of the predictive mean, which is consistent with mild under-confidence from averaging many stochastic passes. The narrow spread shows that ensemble members share a similar logit scale and calibration, which is desirable for stability.

C. COMPUTE-PERFORMANCE TRADEOFFS

Fig. 11 shows the expected tradeoff. Training time scales with the number of models: single-model baselines (MCD, MCSD, MC-FTDB, DUQ, and SNGP) are light, while ensemble methods (E-Dropout, EMCD, LEMC-FTDB) are correspondingly heavier. Inference time scales with the total number of stochastic evaluations: E-Dropout, DUQ, and SNGP are the fastest ($T = 1$), EMCD and LEMC-FTDB are the slowest as they evaluate $M \times T$ passes. LEMC-FTDB is slightly slower than EMCD due to the learned aggregator.

We evaluated ensemble configurations with $M \in \{5, 10, 15\}$ models and $T \in \{10, 50, 100, 150, 200\}$ stochastic passes per model on both EEGNet and EEG-ITNet. We found that M is the primary lever: moving M from 5 to 15 yields ≈ 1 –2 pp gains in correctness, ≈ 0.01 – 0.02 in κ , and ≈ 0.03 – 0.05 lower NLL, alongside reduced predictive entropy and increased MI—i.e., sharper and more epistemically diverse predictions. In contrast, benefits from larger T saturate quickly: once $T \geq 50$ (certainly $T \geq 100$), improvement plateaus; only $T = 10$ is consistently weaker (higher NLL, slightly lower MI). Calibration remains stable across settings ($ECE \approx 0.04$ – 0.05). Practically, $M \approx 10$ with $T \in [50, 100]$ offers a strong compute-efficient

TABLE 4. Ablation study of LEMC-FTDB components with EEGNet.

Mean Metrics	No Weight, No Temp	Weights, No Temp	No Weight, with Temp	Weights and Temp
A (%)	80.32 ± 10.11	80.17 ± 10.44	80.25 ± 10.21	80.40 ± 10.32
κ	0.74 ± 0.13	0.74 ± 0.14	0.74 ± 0.14	0.74 ± 0.14
$\mathbb{H}_{\text{pred}}$	1.37 ± 0.23	0.83 ± 0.29	1.14 ± 0.27	0.80 ± 0.28
MI	0.31 ± 0.04	0.27 ± 0.10	0.49 ± 0.10	0.45 ± 0.15
ECE	0.19 ± 0.03	0.05 ± 0.01	0.12 ± 0.03	0.04 ± 0.01
NLL	0.63 ± 0.21	0.50 ± 0.24	0.56 ± 0.22	0.50 ± 0.24
Brier	0.33 ± 0.13	0.27 ± 0.13	0.30 ± 0.13	0.27 ± 0.13
AUC	0.77	0.82	0.78	0.81
ROC	± 0.08	± 0.07	± 0.08	± 0.08

TABLE 5. Ablation study of LEMC-FTDB components with EEG-ITNet.

Mean Metrics	No Weight, No Temp	Weights, No Temp	No Weight, with Temp	Weights and Temp
A (%)	82.29 ± 9.17	82.91 ± 8.72	82.22 ± 9.30	82.48 ± 8.82
κ	0.76 ± 0.12	0.77 ± 0.12	0.76 ± 0.12	0.77 ± 0.12
$\mathbb{H}_{\text{pred}}$	1.33 ± 0.28	0.79 ± 0.28	1.13 ± 0.30	0.78 ± 0.26
MI	0.36 ± 0.05	0.33 ± 0.11	0.56 ± 0.13	0.52 ± 0.18
ECE	0.20 ± 0.04	0.05 ± 0.01	0.14 ± 0.04	0.05 ± 0.02
NLL	0.60 ± 0.21	0.46 ± 0.22	0.54 ± 0.21	0.46 ± 0.21
Brier	0.31 ± 0.12	0.25 ± 0.12	0.28 ± 0.12	0.25 ± 0.12
AUC	0.80	0.83	0.81	0.83
ROC	± 0.06	± 0.07	± 0.07	± 0.07

operating point, while $M = 15$ provides only incremental improvements if budget allows.

D. ABLATION STUDY

Tables 4 and 5 show the results of the ablation study conducted on the components of our LEMC-FTDB framework. Across both backbones and all subjects, the learned weights are the main driver of probabilistic quality and reliability: they reduce NLL and Brier from $\approx 0.60/0.31$ (No Weight, No Temp) to 0.50/0.27 on EEGNet and 0.46/0.25 on EEG-ITNet (Weights, No Temp), and also lift misclassification-detection AUC ROC (EEGNet: 0.77 to 0.82; EEG-ITNet: 0.80 to 0.83), indicating better separation of correct versus wrong trials. In addition, using the learned weights sharply reduces calibration error—ECE drops (EEGNet: 0.19 to 0.05; EEG-ITNet: 0.20 to 0.05)—with the low variability suggesting that calibration gains are consistent.

Per-model temperature scaling (No Weight, with Temp) contributes differently: it increases epistemic diversity—i.e., higher MI observed (EEGNet: 0.31 to 0.49; EEG-ITNet: 0.36 to 0.56)—and lowers predictive entropy (EEGNet: 1.37 to

1.14; EEG-ITNet: 1.33 to 1.13), but alone it does not match the learned weights on NLL/Brier.

Combining both (Weights and Temp) yields the best calibration overall (ECE = 0.04 on EEGNet; 0.05 on EEG-ITNet) while preserving the strong NLL/Brier of the learned-weighted case, and the increased MI and lower entropy of the per-model temperature scaling. In addition, combining both yields the top or tied accuracy with stable κ and similar SDs across settings. Overall, learned weights improve scoring and error detection, while per-model temperatures enhance diversity and sharpness; together they deliver the most calibrated predictions without sacrificing accuracy.

Compared with a uniform average after temperature scaling, the learned-weight aggregator provides nontrivial improvements in probabilistic quality—notably lower NLL, ECE, and Brier—without sacrificing correctness and κ . Per-model temperature scaling is a low-cost, principled way to correct member-specific logit scales and is most beneficial under uniform averaging; however, when the aggregator is used, a single shared temperature may stand as a lightweight alternative in compute-constrained settings, with only modest degradation primarily in ECE and NLL.

E. REAL-WORLD UTILITY

We used threshold-free metrics, but in real-world deployment a fixed predictive-entropy threshold must be set depending on the target application, to translate confidence into action. In practice, the entropy threshold θ is fixed on a small calibration split by either: 1) an ROC criterion (e.g., balancing TPR–FPR, or enforcing $\text{FPR} \leq 5\%$ for false accepts), or 2) a risk–coverage target (choose θ to achieve, for example, $\leq 10\%$ misclassification among accepted trials or $\leq 60\%$ coverage). At run time, windows with $\mathbb{H}_{\text{pred}}(x) \leq \theta$ are accepted (command emitted) and those with $\mathbb{H}_{\text{pred}}(x) > \theta$ are rejected; a light 3-window smoothing and a small hysteresis margin avoid flicker. This implements a mini, real-time selective prediction policy that exposes the same tradeoff seen in our ROC and risk–coverage curves, demonstrating clinical utility: for the same safety target, LEMC-FTDB accepts more trials and with fewer misclassification. Temperature scaling keeps accepted probabilities well calibrated (low ECE), and a simple MI drift monitor can trigger recalibration if distribution shift is detected.

VI. CONCLUSION

This work introduced LEMC-FTDB—a model-agnostic Learned-weight Ensemble of MC FT-DropBlock networks with Per-Model Temperature scaling—which augments CNN-based EEG decoders with trustworthy uncertainty estimates, centered on predictive entropy. Across the EEGNet and EEG-ITNet backbones using the BCI Competition IV 2a dataset, LEMC-FTDB consistently achieved better probability quality (lower NLL and Brier), improved calibration (lower ECE), stronger misclassification detection, and

superior risk–coverage profiles compared to single-model or uniform-ensemble baselines while maintaining superior correctness and Cohen’s kappa. These findings indicate that combining structured MC masking, per-model temperatures, and learned aggregation yields uncertainty estimates that are both informative and actionable for clinical BCI workflows.

Future work will expand the experiment to datasets with more subjects for more diversity and statistical power in various BCI paradigms. One may explore domain adaptation to handle nonstationary EEG, incorporate self-supervised pretraining to further enhance uncertainty fidelity, and integrate SWAG or MC BatchNorm within our FT-DropBlock pipeline to test whether weight-posterior sampling yields additional gains. In addition, on-device optimization will further streamline entropy-guided selective prediction in routine practice (e.g., FP16/INT8 quantization, pruning, and adaptive $M \times T$ budgeting) to reduce latency and power; alongside engineering choices such as batching windows and using GPU/edge accelerators to fit real-time budgets.

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